

and storage of tropical forests. By using trees as scaffolds, lianas climb to the canopy where they shade tree leaves, restricting tree growth and even killing trees by their tangling presence.

Concerns are also growing that climate change is helping fuel the rapid growth of these vines, and deforestation is leaving space in forests for lianas to flourish.

To understand the potential problems being caused by lianas, experts need to see how quickly these plants are growing and how much of the forest canopy they already cover.

Liana infestation of trees is traditionally measured from the ground, which can be labour-intensive, time-consuming and difficult, particularly for those in the upper canopy layers. Now, Catherine Waite and her colleagues from School of Geography at University of Nottingham, have pioneered a technique that will make the study of lianas much more affordable and efficient, as well as more accurate for trees in upper canopy layers.

Using lightweight unmanned aerial vehicles (drones), Waite was able to capture aerial images of the trees, which enabled her to accurately see the extent of the liana infestation from above the canopy.

New T-wave detector uses waves of the electronic sea in graphene

Any system for wireless data transfer relies on electromagnetic wave sources and detectors, but these are not available for every kind of waves. Existing sources of terahertz radiation, which occupies a middle ground between microwaves and infrared (IR) light, consume too much power or require intense cooling. Yet, T-waves could potentially enable faster Wi-Fi, new

methods of medical diagnostics and studies of space objects using radio telescopes.

Researchers from Russia, Great Britain, Japan and Italy have come up with a solution for the long-standing problem of resonant T-wave detection. They created a photodetector made of bilayer graphene encapsulated between crystals of boron nitride, and coupled to a terahertz antenna. In this sandwich structure, impurities are expelled to the exterior of the graphene flake, enabling plasmons to propagate freely. The graphene sheet confined by metal leads forms a plasmon resonator, and the bilayer structure of graphene enables wave velocity tuning in a wide range.

The team has developed a compact terahertz spectrometer—several microns in size—with the resonant frequency controlled via voltage tuning. The physicists have also shown the potential of their detector for fundamental research. By measuring current in the detector at various frequencies and electron densities, plasmon properties can be revealed.

Dmitry Svintsov, head of Laboratory of 2D Materials for Optoelectronics at Moscow Institute of Physics, says, “Our device doubles up as a sensitive detector and a spectrometer operating in the terahertz range. It is also a tool for studying plasmons in two-dimensional materials. All of these things existed before but took up a whole optical table. We packed the same functionality into a dozen micrometres.”

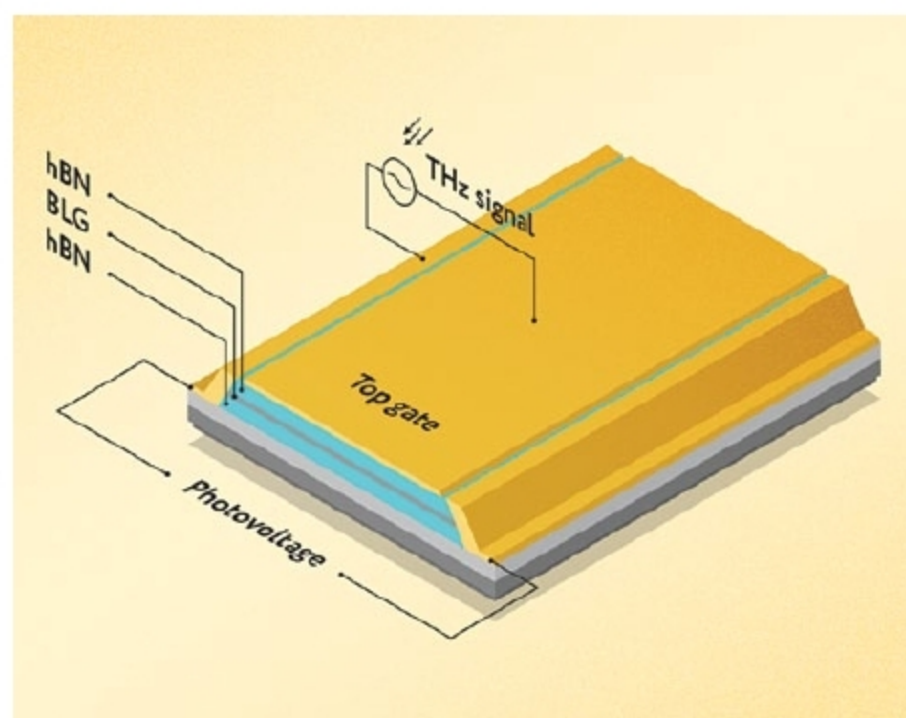
An end-to-end imitation learning system for speed control of autonomous vehicles

Researchers at Valeo, a tech company specialising in automotive innovation, have developed an end-to-end imitation learning system for car speed control. According to a paper pre-published on *arXiv*, their approach uses a neural network with long short-term memory (LSTM), a type of recurrent neural network (RNN) that can learn long-term dependencies.

The new system employs an artificial neural network (ANN) that relies on deep learning techniques. The network is fed demonstrations of a human being operating a car that are taken from a frontal camera and, hence, closely resemble what the person was seeing while driving.

The neural network is then trained to imitate the driver's actions, particularly focussing on reproducing the car's current speed. For instance, when an input image contains a 50kph speed limit panel, the network ensures that the car does not go faster than 50kph.

After training its neural network, the team tested it in a simulation environment and then integrated it inside a real car, evaluating its performance on a challenging test track. The team found that the system effectively reacted to complex situations, controlling the speed of the car where necessary (such as slowing down at traffic cones and sharp turns, stopping at barriers and when approaching warning signs).



Transistor channel, made of bilayer graphene (BLG), is sandwiched between two crystals of hexagonal boron-nitride (hBN). This structure is placed on an oxidised silicon substrate (shown in grey). Two sleeves of a terahertz antenna are connected between the source and the top gate, that is, left and top electrodes (shown in gold). Signal voltage is read between the source and drain terminals, that is, right and left electrodes (Credit: MIPT)